

Using NIW observations to assess mixed layer parameterizations: a case study in the tropical Atlantic

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Key Points:

- Observations of inertial oscillations are used to evaluate the performance of two vertical mixing schemes in two high-resolution models.
- Both the K-profile parameterization (KPP) and the Turbulent Kinetic Energy (TKE) closure underestimate the NIW-induced mixing.
- Near-inertial kinetic energy biases are sensitive to the vertical mixing parameterization.

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Abstract

Tropical sea surface temperature (SST) biases can cause atmospheric biases on global scales, hence SST needs to be represented well in climate models. A major source of uncertainties is the representation of turbulent mixing in the oceanic boundary layer, or mixed layer (ML). In the present study we focus on near-inertial wave (NIW) induced mixing. The performance of two mixing schemes, turbulent kinetic energy (TKE) and K-profile parameterization (KPP), is assessed at two sites (11.5°N, 23°W and 15°N, 38°W) in the tropical Atlantic. At 11.5°N, turbulence observations (eddy diffusivities, shear and stratification) are available for comparison. We find that the schemes differ in their representation of NIWs, but both under-represent the observed enhanced diffusivities below the observed ML. However, we find that the models do mix below the ML at 15°N when a storm passes nearby. The near-inertial oscillations remain below the mixed layer for the following 10 days. Near-inertial kinetic energy (NIKE) biases in the models are not directly correlated with the wind speed, the MLD biases, or the stratification at the mixed layer base. Instead, NIKE biases are sensitive to the vertical mixing scheme parameterization. NIKE biases are lowest when the KPP scheme is used.

Plain Language Summary

The surface temperature of the ocean is highly dependent on the depth of the mixed layer (ML), the uppermost layer in the water column, where density, temperature and salinity are approximately constant. In climate models, the vertical mixing processes cannot be resolved, and instead they are computed with the use of vertical mixing schemes. We assess how well two of such schemes can represent the mixing induced by a specific type of ocean waves, near-inertial waves (NIWs). We compare recent observations of turbulent mixing induced by NIWs in the tropical Atlantic with numerical simulations that resolve storms. Our results show that the models are able to reproduce the observed NIWs, but underestimate their mixing and amplitude. Our analysis also shows that NIWs are a driver of mixing below the uppermost ocean layer in the models. The strength of the near-inertial currents is sensitive to the vertical mixing parameterization.

1 Introduction

The tropical ocean is one of the main energy sources for the tropical and extra-tropical atmosphere and even small sea surface temperature (SST) anomalies there can lead to a restructuring of the global climate (Barsugli & Sardeshmukh, 2002; Jochum & Potemra, 2008). Weather forecasting and climate prediction therefore rely on accurate representation of the tropical SST. A large source of bias could be the vertical mixing (Foltz et al., 2003).

Diapycnal mixing in the ocean has been studied extensively, as it is theorised to be a significant driver of the meridional overturning circulation (Munk & Wunsch, 1998). Vertical turbulent mixing in the ocean interior is mainly driven by internal wave breaking powered by winds and tides (Wunsch & Ferrari, 2004). Near-inertial waves (NIWs) are a dominant mode of high-frequency variability in the ocean (Garrett & Munk, 1975; Fu, 1981). They can be generated by a range of processes, such as oceanic Lee waves and wave-wave interactions (Alford et al., 2016), but their most notable and well-recorded driver is the surface wind stress (e.g., D’Asaro, 1985). Due to their ubiquity in the ocean and the potential for driving diapycnal mixing, a multitude of estimates for the fraction of the wind power input (WPI) transferred into deep ocean by NIWs exist. While the estimates vary in magnitude, a common conclusion is that the majority of the NIW energy is dissipated in the surface ocean. Zhai et al. (2009), for example, show in a model study that almost 70% of the near-inertial WPI is dissipated in the top 200m.

It is well established that NIWs are important in deepening of the surface mixed layer (ML) in the tropical ocean. Strong vertical shear across the mixed layer base induced by near-inertial currents leads to elevated mixing and vertical diffusive heat flux that cools and subsequently deepens the mixed layer (e.g. Large & Crawford, 1995). Jochum et al. (2013) find that by accounting for near-inertial currents in the coarse resolution CCSM4 model, the tropical rainfall bias is reduced through deepening of the ML and changing the SST gradients. NIW-induced mixing is also the dominant process which contributes to SST cooling during and after the passing of tropical cyclones (e.g., Price, 1981; Greatbatch, 1983; Vincent et al., 2012).

The database of direct measurements of NIW-induced mixing has been expanding over the past two decades (e.g., Alford & Gregg, 2001; Hummels et al., 2020; Brizuela et al., 2023), but is still limited. Large scale studies of NIW-induced mixing are therefore based on numerical models of varying complexity (e.g., Rimac et al., 2016; Kuwano-Yoshida et al., 2017; Raja et al., 2022). In climate models, the strength of NIWs is often underestimated because their main generation mechanism is a rapid change of wind speed across the fronts of a storm system, something that is often not properly resolved in ESMs (Rimac et al., 2013). The amplitude of wind-driven NIWs is highly dependent on the depth of the mixed layer (MLD) (Pollard & Millard, 1970), and their rate of dissipation is primarily dependent on ocean stratification (Pollard, 1970).

The biases in surface ocean MLD remain an unsolved problem in ocean modeling (Huang et al., 2014). One issue is that the depth of the mixed layer is a highly spatially and temporally variable field (de Boyer Montégut et al., 2004). While ocean models are able to simulate realistic MLD values, it remains a question whether local mixing processes are accurately represented. Our study attempts to bridge this gap by evaluating the ability of two commonly used mixing schemes, the K-profile parameterization (KPP, Large et al., 1994) and the turbulent kinetic energy closure scheme (TKE, Gaspar et al., 1990), to reproduce NIW-induced mixing in the tropical Atlantic. We use two forced, eddy ocean models: the Ocean component of the ICOSahedral and Nonhydrostatic weather and climate model (ICON-O, Korn et al., 2022) and the Finite Element/volumeE Sea ice-Ocean Model (FESOM, Danilov et al., 2017; Scholz et al., 2019). Furthermore, we assess the sensitivity of the NIW-induced mixing to the TKE scheme parameteriza-

tion by conducting five TKE sensitivity experiments. We present the results of this evaluation at two sites in the tropical Atlantic.

The text is structured as follows: we motivate the choice of sites (section 2.1), specify the setup of the sensitivity studies (section 2.2) and the data selected to evaluate the model performance (section 2.3); the atmospheric conditions at the sites are compared to model forcing (section 2.4). We present an overview of the MLD bias in the tropical Atlantic (section 3.1), modelled NIW characteristics (section 3.2), their mixing signature (section 3.3), and the sensitivity of the near-inertial kinetic energy to the vertical mixing schemes (section 3.4). Following a summary and discussion (section 4), the main conclusions of the study are listed in section 5.

2 Methods

2.1 Choice of study period and locations

A study by Hummels et al. (2020) (hereafter referred to as HDRF20) documents a strong near-inertial wave observed during a research cruise in the tropical Atlantic. The vessel-mounted Acoustic Doppler Current Profiler (vmADCP) measurements show a clear signal of the wave at 11°N, 21°W between 13th and 15th of September, 2015, as the tropical storm Ida develops (Cangialosi, 2015). The shear induced by the wave cooled the mixed layer at the rate of 244 Wm^{-2} . The observations provide a complete set of parameters required to investigate the NIW-induced mixing in the models, therefore we pick the years 2014/15 as the period of the ocean simulations.

The Prediction and Research Moored Array in the Tropical Atlantic (PIRATA, Bourlès et al., 2019) offers a range of long-term measurements of ocean currents and state. In the Atlantic, the near-inertial velocity variance is highest between 7.5°S and 12.5°N (Elipot & Lumpkin, 2008). In the vicinity of this latitude band, high frequency current and co-located wind observations are available in 2014/15 at two PIRATA moorings off the equator: 11.5°N, 23°W; 15°N, 38°W. Coincidentally, Ida also passes across this second mooring, and the two periods of NIW-induced mixing are the focus of the present study. Throughout the article, they are referred to as the 11.5°N and the 15°N site, respectively.

2.2 Setup of sensitivity experiments

We use two models for the study: the Ocean component of the ICOSahedral and Nonhydrostatic weather and climate model (ICON-O, Korn et al., 2022) and the Finite Element/volumE Sea ice-Ocean Model (FESOM, Danilov et al., 2017; Scholz et al., 2019). Both of them are forced with hourly ECMWF Reanalysis product (ERA5, Hersbach et al., 2020). FESOM spin-up is initialized from the Polar Science Hydrographic winter Climatology (PHC3.0, updated from Steele et al., 2001) in 2009. Both the base run and the sensitivity runs start from rest. ICON-O spin-up is initialized with Ocean Reanalysis System 5 (ORAS5, Zuo et al., 2019) in 2010. The simulation branches out into control and sensitivity experiments in 2014. The models run for 2 simulated years after the spin-up with a temporal output frequency of 3h.

The models run on a triangular mesh grid, and the output is interpolated on a regular 0.1° grid in post-processing. There are 40 vertical levels in the top 200m, with monotonously increasing distance between the layers: 2m at the surface and 10.7m at the bottom. Due to constraints on available output storage space, the simulation domain covers the whole global ocean down to the sediment, but only the upper 200m are saved. FESOM runs on a B-grid of 13km resolution in the tropical Atlantic and 50km resolution in the rest of the ocean. ICON-O runs on a C-grid of approximately 10km horizontal resolution globally. We use two widely employed vertical mixing schemes for the analysis: the turbu-

lent kinetic energy (TKE, Gaspar et al., 1990) and the K-profile parameterization (KPP, Large et al., 1994).

TKE is a parameterization where the turbulent vertical fluxes are determined from the vertical property gradients. Eddy diffusivities are dependent on TKE:

$$K_m = c_k l_k \bar{\epsilon}^{1/2}, \quad K_h = K_s = K_m / P_{rt}, \quad (1)$$

where l_k is a mixing length, c_k is a constant and $\bar{\epsilon}$ is the TKE. The momentum diffusivity, K_m , is related to the heat and salinity diffusivities, K_h and K_s , through the Prandtl number. The constant c_k scales the contribution of turbulent kinetic energy to mixing. The TKE itself is computed by a prognostic equation in the model. It can be enhanced by external forcing, such as the winds, or internal waves and currents producing shear. Aside from the TKE dissipation term, TKE sinks include vertical TKE flux and stable stratification.

KPP is a non-local closure parameterization. Like TKE, KPP utilizes the concept of eddy diffusivities for determining the turbulent vertical fluxes. The diffusivities no longer depend on TKE, but have a vertical profile dependent on the depth of the boundary layer h , which is determined using the bulk Richardson number:

$$Ri_b(d) = \frac{(B_r - B(d))d}{|\mathbf{v}_r - \mathbf{v}(d)|^2 + v_t^2(d)}, \quad (2)$$

where B is buoyancy, $\mathbf{v} = (u, v, w)$ is the mean velocity, and the subscript r denotes near-surface values; v_t is an additional term, the turbulent velocity shear, which ensures that entrainment is independent of stratification N . The depth of the boundary layer h is then the smallest d for which eq. (2) is less than a critical bulk Richardson number, Ri_c . Below the boundary layer, the ocean vertical mixing in KPP is parameterized as a superposition of three effects: instability due to resolved shear, internal wave breaking, and double diffusion. The former is based on the local gradient Richardson number, and the diffusivities arising from internal wave breaking are set to constant values (typically $10^{-4} \text{m}^2 \text{s}^{-2}$ for momentum and $10^{-5} \text{m}^2 \text{s}^{-2}$ for tracer diffusivity).

The vertical mixing schemes are implemented in the models with the Community Ocean Vertical Mixing package (CVMix, Griffies et al., 2015; Van Roekel et al., 2018). The evaluation of the schemes can be found in Korn et al. (2022), Gutjahr et al. (2021) (ICON-O), and Scholz et al. (2022) (FESOM). In the control runs, the vertical mixing is parameterized with the TKE scheme. The minimum background TKE is 10^{-6}J kg^{-1} , with no minimum diffusivity imposed, and $c_k = 0.2$. The sensitivity experiments in both of the models include TKE runs with $c_k = 0.1$ and $c_k = 0.3$, as well as a simulation done with KPP. During initial runs, we noticed a low diffusivity band developing at the bottom of the mixed layer, leading to introducing two additional sensitivity runs: one with minimum background TKE of 10^{-5}J kg^{-1} ; and another with a minimum background K_m of $10^{-4} \text{m}^2 \text{s}^{-1}$ and a minimum background K_h of $10^{-5} \text{m}^2 \text{s}^{-1}$. The two KPP experiments – ICON_kpp and FESOM_kpp – are both run with the critical Richardson number $Ri_c = 0.3$. The minimum background K_h and K_m values are set as in ICON_kappa. In ICON_kpp, the mixing scheme below the ocean boundary layer is PP (Pacanowski & Philander, 1981), while in FESOM_kpp, it is KPP at both the surface and in the interior of the ocean. Table 1 outlines the specifications of the experiments.

2.3 Observations used for model evaluation

The modelled mixing in the tropical Atlantic is assessed both by comparison to MLD climatology and by investigating the local effect of NIW mixing at the two selected sites. The latter requires model forcing evaluation to ensure that major discrepancies between modelled and observed NIW properties originate from the mixing parameterizations, rather than biased surface boundary conditions. We use three main data sources for these pur-

Table 1. Overview of the experiments.

Name	Mixing scheme	c_k value	Minimum TKE	Minimum κ_H	Minimum κ_M	Ri_c
ICON_ck1	TKE	0.1	10^{-6}	—	—	—
ICON_ck2	TKE	0.2	10^{-6}	—	—	—
ICON_ck3	TKE	0.3	10^{-6}	—	—	—
ICON_kpp	KPP	—	—	10^{-5}	10^{-4}	0.3
ICON_mintke	TKE	0.2	10^{-5}	—	—	—
ICON_kappa	TKE	0.2	10^{-6}	10^{-5}	10^{-4}	—
FESOM_ck1	TKE	0.1	10^{-6}	—	—	—
FESOM_ck2	TKE	0.2	10^{-6}	—	—	—
FESOM_ck3	TKE	0.3	10^{-6}	—	—	—
FESOM_kpp	KPP	—	—	10^{-5}	10^{-4}	0.3

poses: ship based climatology, the PIRATA array, and observations from the R/V Meteor cruise M119.

For the MLD, we use the IFREMER climatology (de Boyer Montégut et al., 2004), which is a global dataset computed from profiles collected between 1941 and 2008. The MLD is calculated using the criterion $\Delta T = \pm 0.2^\circ C$ from the in-situ temperature at 10m below the surface. The annual climatology set contains spatially averaged median MLD values, computed from profiles grouped into $2^\circ \times 2^\circ$ grid boxes. Additionally, we use the enhanced PIRATA dataset (Foltz et al., 2018) to compare the time series of the modelled and observed MLD at the two PIRATA sites. In ePIRATA, Argo data is used to construct vertical temperature and salinity profiles from the mooring observations; the MLD time series is then calculated from the criterion of $\Delta \rho = 0.12 \text{ kg m}^{-3}$ relative to 1m depth, which is approximately equivalent to $\Delta T = -0.35^\circ C$.

A range of PIRATA data is used to validate the forcing and the NIW response. We note that the data from the PIRATA moorings are assimilated into the ERA5 product as a part of the DRIBU dataset (Hersbach et al., 2020). In the forcing comparison, we use the following: wind vector data measured at 4m above the sea surface (accuracy of magnitude $\pm 2\%$ and direction $\pm 3.4^\circ$); longwave and shortwave radiation data measured at 3.5m above the sea surface ($\pm 1\%$ and $\pm 2\%$, respectively); and total precipitation data ($\pm 0.4 \text{ mm hr}^{-1}$). The wind vector at 10m is computed following the logarithmic wind profile assuming neutral stability (Oke, 2002):

$$\mathbf{U}_{10} = \mathbf{U}_4 \frac{\ln(10/z_0)}{\ln(4/z_0)}, \quad (3)$$

where $z_0 = 2 \times 10^{-4} \text{ m}$ is the roughness length and $\mathbf{U} = (U, V)$ is the wind vector. For NIW response validation, we use high frequency ocean current ($\pm 0.5 \text{ cm s}^{-1}$) and temperature data ($\pm 0.002^\circ C$).

Finally, for the comparison of the mixing parameters, we use the microstructure and vmADCP measurements from the M119 cruise (Brandt et al., 2017; Fischer, 2020). The NIW observations published by HDRF20 were recorded between the 13th and 15th of September. The shear ($Sh^2 = (\partial u / \partial z)^2 + (\partial v / \partial z)^2$) is computed from the vmADCP current measurements, which span the depths 17m to 800m, with 8m spatial resolution. The shear values are averaged between the 13th and 15th of September (with cutoff at midnight). The 25 microstructure profiles were recorded between the 13th of September, 10:00 UTC and the 14th of September, 06:00 UTC, and have the vertical resolution of about 0.5m. The 95% confidence intervals of N^2 and Sh^2 are estimated using the bias-corrected and accelerated (BCa) bootstrapping algorithm. Eddy diffusivity values across the 25 profiles are collected in 10m depth bins and averaged. The 95% confidence intervals of K_ρ are estimated using error propagation from Schafstall et al. (2010).

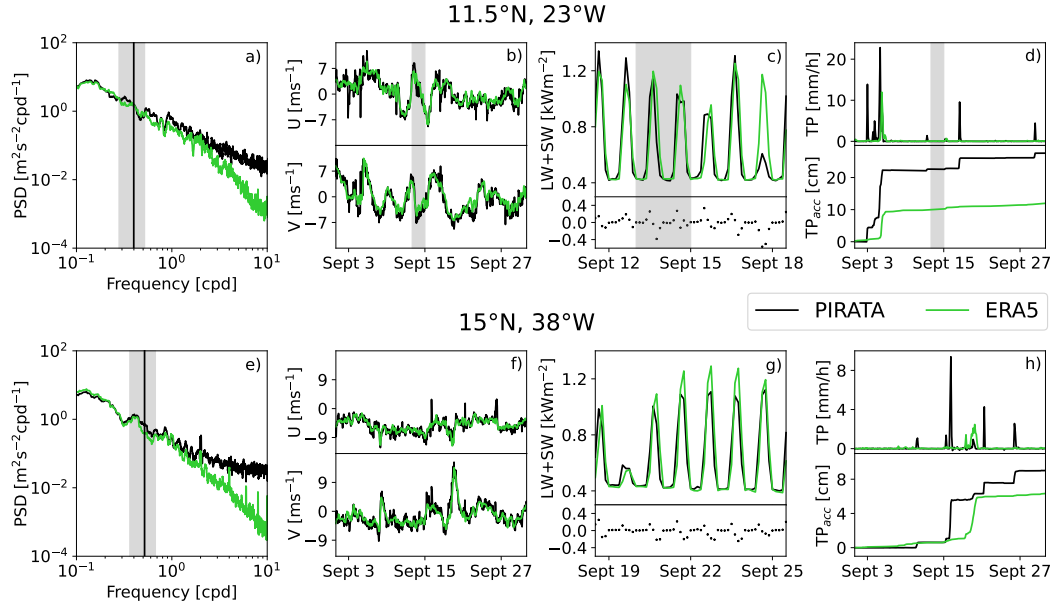


Figure 1. The comparison of the ERA5 forcing (green) to PIRATA mooring data (black) at 11.5°N and 15°N in September 2015. Subplots a) and e) show the power spectral density (PSD) of the wind speed time series spanning from January to October 2015, the black vertical lines indicate the local Coriolis frequency f , and the shaded region spans from $0.7f$ to $1.3f$; b) and f) show the zonal and meridional wind velocities (U , V); c) and g) show the sum of the long- and shortwave radiation (LW+SW) at the time of the NIW events; the plots below indicate the residual (PIRATA-ERA5); d) and h) show the total precipitation (TP) and accumulated precipitation (TP_{acc}). In b), c), and d), the shaded region indicates the duration of HDRF20 observations.

2.4 Comparison of forcing to PIRATA data

The significant factors for NIW generation and dissipation are the wind stress, mixed layer depth, and stratification. The relevant local forcing parameters to evaluate are therefore wind velocities, radiation, and rainfall. The agreement between the ERA5 forcing and the local measured PIRATA climate is evaluated and presented in Figure 1. The mixed layer depth is evaluated in section 3.1.

NIWs are forced by the wind stress. In comparison to the direct PIRATA observations, the power spectral density (PSD) of the ERA5 wind speed at the two sites in 2015 shows a clear difference at periods of hours or less (Fig. 1a and 1e). This difference is expected, because at scales close to grid scale viscosity is designed to remove energy to suppress numerical instability (Jochum et al., 2008). Within the near-inertial band, the wind energy is 10% lower at the 11.5°N site and 27% lower at the 15°N site in ERA5 relative to PIRATA. The wind velocity time series in Fig. 1b and 1f confirm the coherence of the wind forcing. We can therefore expect coherence in the observed and simulated NIW signal, but lower amplitude due to the discrepancy in wind power input.

The shortwave and longwave series reveal a diurnal cycle bias of $200\text{--}300\text{Wm}^{-2}$ (Fig. 1c and 1g). This bias originates from the shortwave radiation data. ERA5 tends to show higher daily radiation values compared to PIRATA, especially at the 15°N site. Considering that the analysed NIWs were associated with the passing of a storm, these biases likely arise from discrepancies between in-situ and modelled cloud cover. Additionally, Foltz et al. (2013) report that some of the PIRATA moorings can show a bias in SW mea-

Table 2. Mean surface salinity (S), salinity difference across the mixed layer (ΔS across the ML), surface potential density (σ_θ) and potential density difference across the mixed layer ($\Delta\sigma_\theta$ across the ML) in the ICON-O model runs and PIRATA mooring data. The means are computed over two time periods (duration of HDRF20 observations, 13th to 15th of September, 2015; and September, 2015) at the 11.5° site and over September, 2015 at the 15°N site. "Across the ML" refers to salinity and potential density values at the surface subtracted from values at a depth d below the mixed layer. At 11.5°N, $d = 20\text{m}$; at 15°N, $d = 40\text{m}$. "TKE average" stands for the average of all the TKE sensitivity runs. In bold, model biases are highlighted. Positive bias indicates the modelled values are higher compared to PIRATA data.

Site	11.5°N	11.5°N	11.5°N	11.5°N	15°N	15°N
Time period	HDRF20	HDRF20	Sept 2015	Sept 2015	Sept 2015	Sept 2015
Surface salinity (S) [psu]						
PIRATA	35.63	-	35.37	-	36.48	-
TKE average	35.37	-0.26	35.49	0.12	36.46	-0.02
KPP	35.27	-0.36	35.32	-0.05	36.57	0.09
ΔS across the ML [psu]						
PIRATA	0.16	-	0.32	-	0.03	-
TKE average	0.31	0.15	0.35	0.03	0.11	0.08
KPP	0.15	-0.01	0.17	-0.15	0.09	0.06
Surface σ_θ [kg m^{-3}]						
PIRATA	22.77	-	22.55	-	23.52	-
TKE average	22.95	0.18	22.81	0.26	23.63	0.11
KPP	22.88	0.11	22.82	0.27	23.82	0.30
$\Delta\sigma_\theta$ across the ML [kg m^{-3}]						
PIRATA	0.51	-	0.83	-	0.73	-
TKE average	1.70	1.19	1.79	0.96	1.24	0.51
KPP	0.49	-0.02	0.55	-0.28	1.08	0.35

surements during boreal summer and fall due to aeolian dust accumulation. The magnitude of the bias can be as large as 50Wm^{-2} , which could account for 25% of the radiation offset between ERA5 and PIRATA. The contribution could be significantly higher during the day, as the reported biases are based on daily averages, including zero values at night.

While the precipitation biases are relatively low at the times the NIWs are measured, long-term precipitation differences can introduce persistent salinity offsets. Therefore, we show the hourly precipitation and the accumulated precipitation over the entirety of September 2015 (Fig. 1d and 1h). Overall, PIRATA registers higher rates of rainfall, but ERA5 overestimates the precipitation at the 15°N site when the storm passes by the mooring on the 19th of September.

Surface salinity fluxes are not consistently impacted by the systematic underestimation of rainfall in ERA5. Table 2 shows the mean salinity and potential density in the ICON-O models and observations. The differences across the mixed layer are computed by subtracting the surface layer values from the values at a depth layer d below the ML: $d = 20\text{m}$ at 11.5°N; $d = 40\text{m}$ at 15°N. The depths are selected based on PIRATA data availability. The means are calculated over the course of September, 2015 at both sites. At 11.5°N, we additionally report the mean biases during HDRF20 observations. "TKE average" is an average of all TKE sensitivity experiments: the temporal means are computed for each run, and an average of those is presented in the table.

Surface salinity is not systematically higher, nor consistently biased at the two sites as a result of reduced rainfall in ERA5. In the tropics, a high salinity layer develops below the fresher ML in the summer. The reduced rainfall in ERA5 would therefore serve to reduce the density gradient across the ML. Instead, the density gradient is generally too high in the models. The surface ocean is too dense regardless of the magnitude of surface salinity biases. There is therefore no clear indication that the models are significantly affected by the underestimated precipitation in ERA5 forcing.

In summary, the wind power input within the near-inertial band is lower by 10% at 11.5°N and by 27% at 15°N in ERA5 compared to PIRATA. Despite the biases in radiative forcing and precipitation, the surface layer salinity and potential density means in the models are all within 1.3% of the observed PIRATA values. We conclude that the ERA5 reanalysis product is sufficiently close to the PIRATA data to force the observed NIWs and their response in the models.

3 Results

3.1 Mixed layer depth in the models

The modelled tropical Atlantic MLD is calculated from the temporally averaged output using the criterion of $\Delta T = -0.2^\circ\text{C}$ relative to temperature at 10m depth. The mean is taken over the summer, defined as 1st of July to the 1st of October. The map of median values in $2^\circ \times 2^\circ$ boxes is computed for all the model runs. In Figure 2, the climatological MLD distributions are shown, together with comparison to the control FESOM run, FESOM_ck2, and the simulation which resembles observations most, FESOM_kpp. The zonal averages of MLD distributions are presented in Fig. 2g for the general overview of the model performances. Local daily MLD time series at 11.5°N and 15°N are shown in Fig. 2c and 2f, respectively, and compared to the ePIRATA reconstructions. The model values are calculated from $\Delta T = -0.35^\circ\text{C}$ relative to 1m depth.

We note that 2015 marked the beginning of the strongest Pacific El Niño event of the 21st century (Santoso et al., 2017). Motivated by this, we check for wind anomalies in the Atlantic. The winds do not diverge significantly from the climatology, but the El Niño event caused reduction in hurricane activity in the tropical Atlantic in 2015 (Stewart, 2016).

The two maps in Fig. 2d and 2e show the shallow MLD bias which spans nearly the entire basin and is present in all experiments. The exception is a band north of the equator where the zonal average of ICON_kpp MLD is about 1.5 times deeper than the IFREMER climatology (Fig. 2g). North of 10°N, all but ICON_kpp modelled MLD profiles are too shallow and fall outside of the IFREMER confidence bound. The ICON-O TKE runs are generally more biased than FESOM, despite identical forcing and mixed layer scheme. The TKE sensitivity experiments do not affect the zonal MLD average significantly; instead we see considerable differences between the two models, and the two mixing schemes. The KPP MLD values are systematically deeper than the TKE based ones.

At the 11.5°N site, the MLD is generally too shallow in all TKE runs, and too deep in the KPP runs, as indicated by the comparison to the ePIRATA data (Fig. 2c). During the HDRF20 observations, ICON_kpp simulates MLD closest to observed. The TKE and KPP runs are not as easily distinguishable at 15°N (Fig. 2d). Here, the MLD before the passing of the tropical storm on the 19th of September varies between 15 and 30m; afterwards, the ML is restratified.

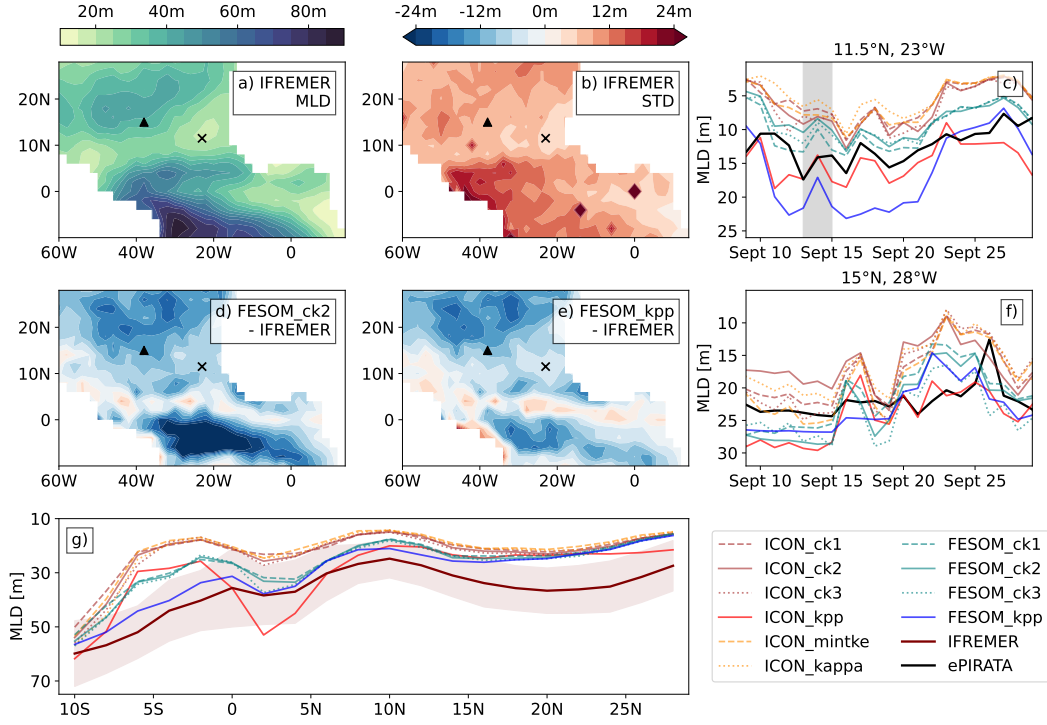


Figure 2. The mixed layer depth in the model runs compared to the IFREMER climatology in the tropical Atlantic. The IFREMER data in a) and the corresponding std in b) show the summer (JAS) average between the years 1941 and 2008, while the model output is the average of summer 2015. The difference maps d) FESOM_ck2-IFREMER and e) FESOM_kpp-IFREMER show the shallow bias which is consistent in all model runs. The 11.5°N site is marked on the maps with an x, and the 15°N site - with a triangle. In g), the zonal averages of all experimental MLD profiles are compared to the climatology. The shading indicates the median std from the raw IFREMER MLD data. c) and f) show the modelled September MLD time series at 11.5°N and 15°N, respectively; the ePIRATA (black) daily MLD values are presented with the estimated error (shading). In c), the gray vertical bar indicates the duration of the HDRF20 observations.

3.2 Near-inertial wave characteristics

The time series of the zonal and meridional velocities at 10m depth is shown in Figures 3a and 3b. A bandpass filter is used to extract the near-inertial signal from the raw data (Fig. 3c and 3d). The Gaussian window envelops frequencies between $0.7f$ and $1.3f$. The filtered velocities, u_i and v_i , are then used to compute the near-inertial kinetic energy:

$$NIKE = \frac{1}{2} \rho_0 (u_i^2 + v_i^2). \quad (4)$$

The NIKE values averaged over September 2015 and the three days of the HDRF20 observations are listed in Table 3. We choose to present the two NIKE diagnostics for the 11.5°N site, as the duration of the HDRF20 observations does not span over a full inertial period. The temporal distribution of NIKE is presented in Figures 3e and 3f.

The models generally underestimate the strength of the near-inertial currents at both sites. At 15°N, the negative bias is higher. About 27% of the lacking NIW energy at this site can be associated with the underestimated wind power input within the near-inertial band in ERA5 (section 2.4). At 11.5°N, despite 10% weaker winds, the mean September

ber NIKE is higher than observed in Icon_ck1, Icon_ck2 and Icon_ck3. In the period of HDRF20 observations, NIWs are too weak in all experiments.

There are key differences in how the NIWs are modelled at the two sites. The waves are in phase at 15°N and out of phase at 11.5°N , and the strength of the near-inertial currents is more sensitive to the vertical mixing scheme at 11.5°N . The difference most likely stems from stratification: at 11.5°N , the potential density gradient across the ML is more than two times higher in the ICON-O TKE models compared to PIRATA (Table 2); at 15°N , the density gradient bias is lower. The magnitudes of the relative MLD biases at the two sites are similar (Fig. 2c and f), but the mixed layer is thinner at 11.5°N . NIW amplitude is inversely proportional to the depth of the mixed layer (D'Asaro, 1985), hence shallow MLD biases at 11.5°N potentially have a larger impact on the strength of NIWs. At both sites, the mean NIKE is sensitive to the amount of turbulent kinetic energy which is used for mixing in the TKE scheme: generally, low c_k values lead to more energetic near-inertial waves. The sensitivity of NIKE bias to the vertical mixing scheme parameterization is further discussed in section 3.4.

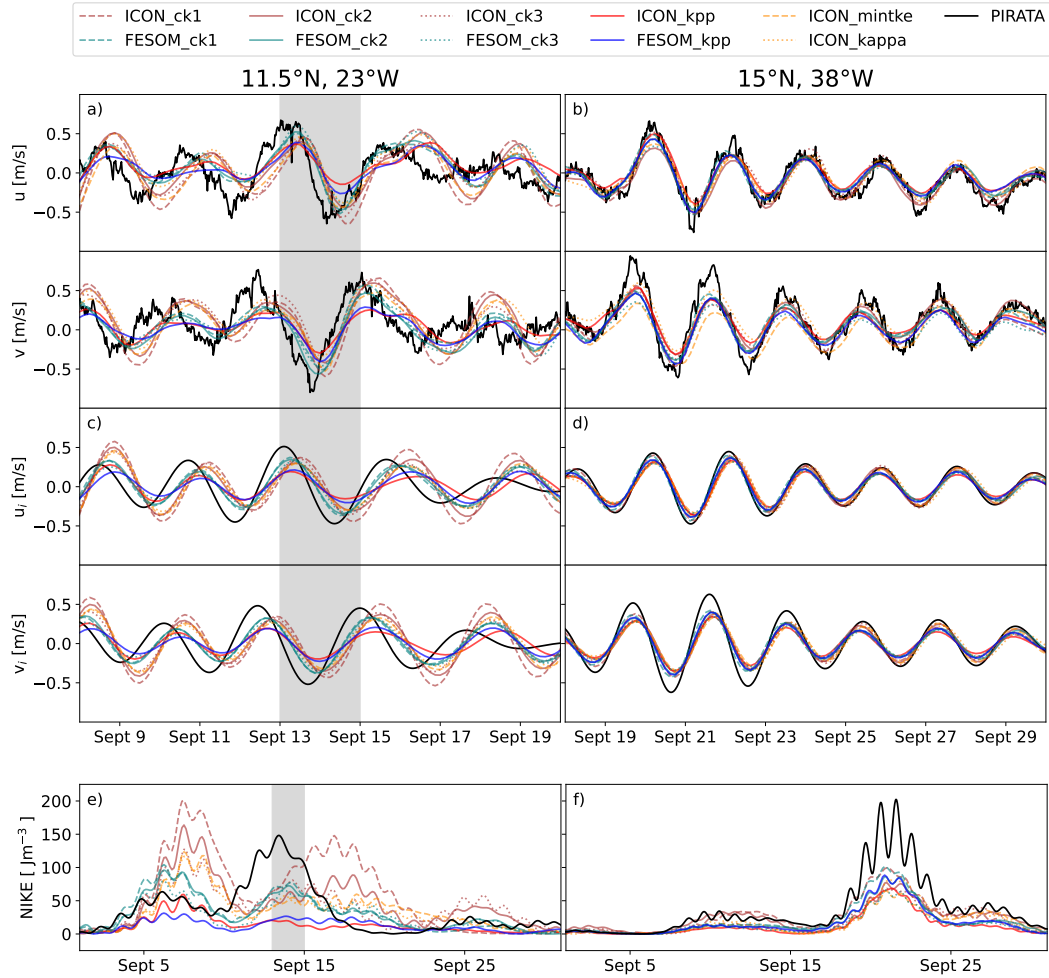


Figure 3. In a) and b), the raw PIRATA mooring velocity data in comparison to modelled velocity output at 11.5°N and 15°N at 10m depth are presented. The bandpassed velocities u_i and v_i are plotted in c) and d). The September NIKE time series for both sites is shown in e) and f). The gray shading indicates the duration of the HDRF20 observations.

Table 3. Overview of the mean NIKE. At 11.5°N, the mean is taken both over the entire September 2015 and over the duration of the HDRF20 observations (13th to 15th of September). The relative difference between the model results and the PIRATA data is highlighted in bold text. Negative values indicate that modelled NIKE is weaker than observed.

Site	11.5°N	11.5°N	11.5°N	11.5°N	15°N	15°N
Time period	HDRF20	HDRF20	Sept 2015	Sept 2015	Sept 2015	Sept 2015
	[Jm ⁻³]	[%]	[Jm ⁻³]	[%]	[Jm ⁻³]	[%]
PIRATA	116.6	-	35.6	-	36.3	-
ICON_ck1	80.8	-30.7	59.2	66.3	25.1	-30.9
ICON_ck2	48.6	-58.3	46.0	29.2	22.7	-37.5
ICON_ck3	76.0	-34.8	42.3	18.8	18.8	-48.2
ICON_kpp	13.2	-88.7	12.9	-63.8	12.5	-65.6
ICON_mintke	40.7	-65.1	35.6	0.0	16.3	-55.1
ICON_kappa	42.9	-63.1	32.8	-7.9	18.2	-49.9
FESOM_ck1	72.6	-37.7	32.1	-9.6	20.6	-43.3
FESOM_ck2	68.0	-41.7	31.6	-11.2	18.1	-50.1
FESOM_ck3	61.6	-47.2	27.3	-23.3	13.9	-61.7
FESOM_kpp	24.1	-79.2	12.3	-65.4	17.5	-51.8

3.3 The mixing signature of NIWs

Detailed turbulence observations are only available for the 11.5°N site. We therefore use the PIRATA temperature data to assess the vertical mixing of heat in the models at both locations (section 3.3.1). The simulated eddy diffusivities, stratification, and shear are evaluated at the 11.5°N site in section 3.3.2. We then present the deep mixing in the models after the storm at 15°N in section 3.3.3.

3.3.1 Temperature

The measured and modelled temperatures at the two sites are compared in Figure 4. In Figures 4f and 4g, the mean September vertical temperature profiles show that in most models, not enough heat penetrates below the mixed layer. The standard deviation from the PIRATA monthly means is indicated by the black horizontal bars. The temperature oscillations below the ML have a near-inertial period, which is reproduced by the models at the 15°N site (demonstrated in Fig. 4k and 4l), but not at 11.5°N (Fig. 4e). At the surface, the diurnal cycle is evident, and it is reproduced by the models. Within the ML, the temperature is approximately constant; the diurnal cycle signal does not reach the bottom of the mixed layer, and this is consistent between most of the models and the PIRATA observations.

ICON_kpp exhibits the best agreement with the observed September mean temperature profile at 11.5°N. The additional ICON TKE runs, ICON_mintke and ICON_kappa, show an improved vertical temperature profile compared to the base run ICON_ck2. The modelled vertical temperature profile is more consistent across the sensitivity studies at the 15°N site. The systematic overestimation of the temperature gradient steepness could be a result of the lacking wind energy in ERA5 forcing. The rapid cooling of the surface occurs when the storm passes by the mooring. All models reproduce the rapid ML cooling and subsequent warming after the storm. Below the ML, the temperature oscillations are present, but have reduced amplitude compared to the observations (Fig. 4k and 4l). Contrary to the 11.5°N site, none of the vertical mixing schemes can be singled out as best performing.

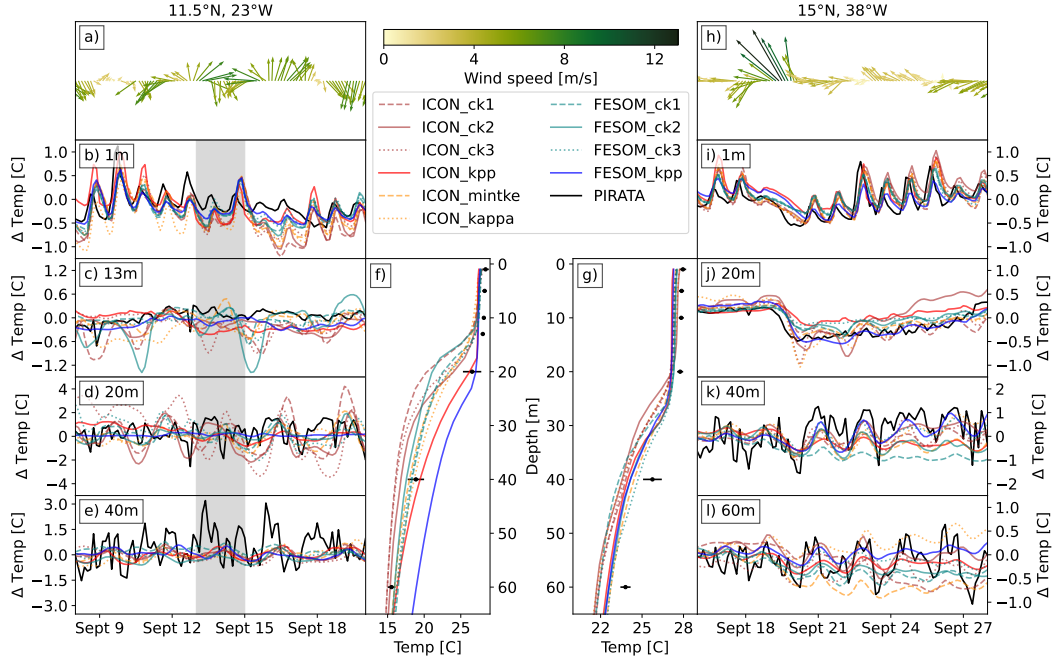


Figure 4. The comparison of measured and modelled temperatures at the two sites. a) and h) show the PIRATA wind vectors. The mean September vertical temperature profiles are given in f) and g); the PIRATA temperature averages for each available depth are indicated with the black points, and the horizontal bars indicate the standard deviation. The time series show departures from September mean temperature at b) 1m, c) 13m, d) 20m, and e) 40m depths for the 11.5°N site; and (i) 1m, (j) 20m, (k) 40m, and (l) 60m depths for the 15°N site. The gray shaded area in b), c), d) and e) and indicates the duration of the HDRF20 observations.

3.3.2 Comparison to microstructure measurements at 11.5°N

The snapshots of the mixing parameters at the 11.5° site are examined in the model sensitivity runs and compared to the HDRF20 data. The temporal locations of the peaks in diffusivities and shear coincide with maximum NIW velocities, which differ across the models due to the NIW phase shift. The snapshots have a vertical structure which persists over time, and therefore we choose to average the profiles to highlight the similarities between the modelled and observed mixing structures. These are presented in Figure 5.

Stratification is well represented in the models beyond the ML. The KPP runs simulate the observed N^2 profile the best, and all models can reproduce the observed stratification below the mixed layer. Above 10m, N^2 is significantly overestimated in all the runs. Shear is systematically underestimated by the models despite coarser vertical resolution of observed horizontal currents. We suspect unresolved internal waves are the reason for this discrepancy (Large et al., 1994).

All models exhibit enhanced mixing above approximately the same depth of 20m. The diffusivity values from the microstructure profiler are only valid below the ML, which is not constant across the 25 measurements. The top bin of microstructure diffusivity data in Fig. 5 is set to depth below 18m, which is the average observed MLD, while the modelled MLD is shallower in the TKE runs. Hence, the TKE simulations exhibit the

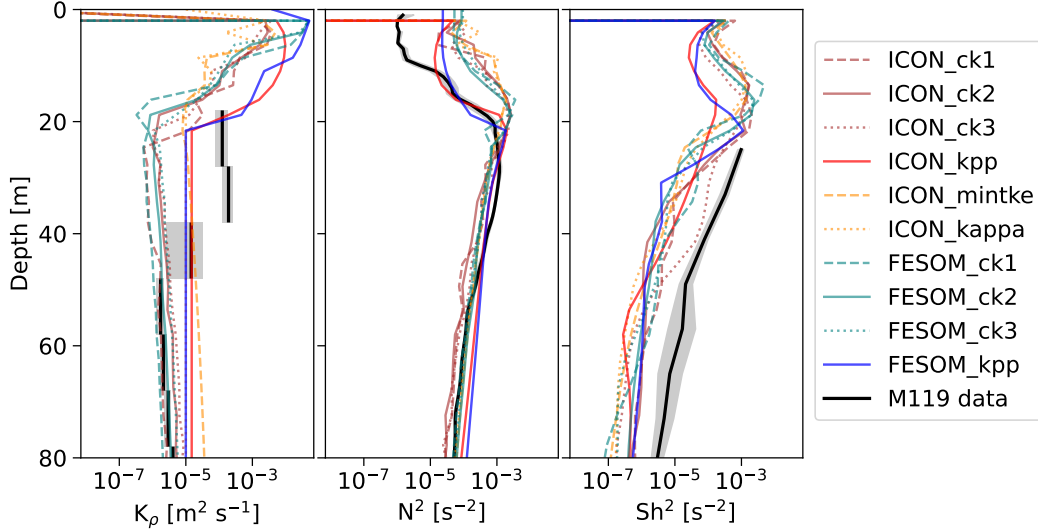


Figure 5. Eddy diffusivities (K_ρ), the buoyancy frequency (N^2) and shear (Sh^2) mean vertical profiles in the models compared to microstructure and vADCP current data from the M119 RV Meteor cruise. The gray shading indicates 95% confidence bounds. The top bin in the diffusivity data is set to 18m, which is the mean depth of the mixed layer at the time of the HDRF20 measurements. Modelled diffusivities within the mixed layer (above 18m) are available and shown.

observed enhanced diffusivities below the simulated ML, but the effect does not reach the same depth due to the thinner mixed layer.

The enhanced diffusivity below 20m is not reproduced by the models. The TKE model runs with no restrictions to background diffusivities match the interior K_ρ values below 50m depth well. Although the vertical resolution of the models is higher than the observations, the interior shear is lower than observed in all of the runs. ICON_ck3 is an exception, coming close to the observed shear at 40m. Despite this local consistency with both the N^2 and Sh^2 values, the run does not exhibit enhanced diffusivity at this depth.

3.3.3 Vertical mixing profile

The lack of deep mixing response to NIWs at 11.5°N puts to question whether such process occurs in the models at all. Due to the proximity and strength of the storm at the 15°N site, we can verify whether mixing below the ML can be simulated.

Snapshots of K_ρ in the model runs reveal that indeed, the enhanced mixing reaches below the recorded ML after the storm passess on the 19th of September (Fig. 6). Diffusivities of $10^{-3} \text{m}^2 \text{s}^{-1}$ reach 40m depth in most of the models. Oscillations at near-inertial frequencies can be seen in the TKE runs; the oscillations persist for over 10 days after the storm passess. The PSD of diffusivities at 53.6m and 58m show clear peaks at the near-inertial frequency.

3.4 NIKE sensitivity to mixing scheme parameterization

There are a few key differences in how NIWs are simulated at the two PIRATA sites: at 11.5°N, the mean NIKE bias is positive for some models, the NIWs are out of phase,

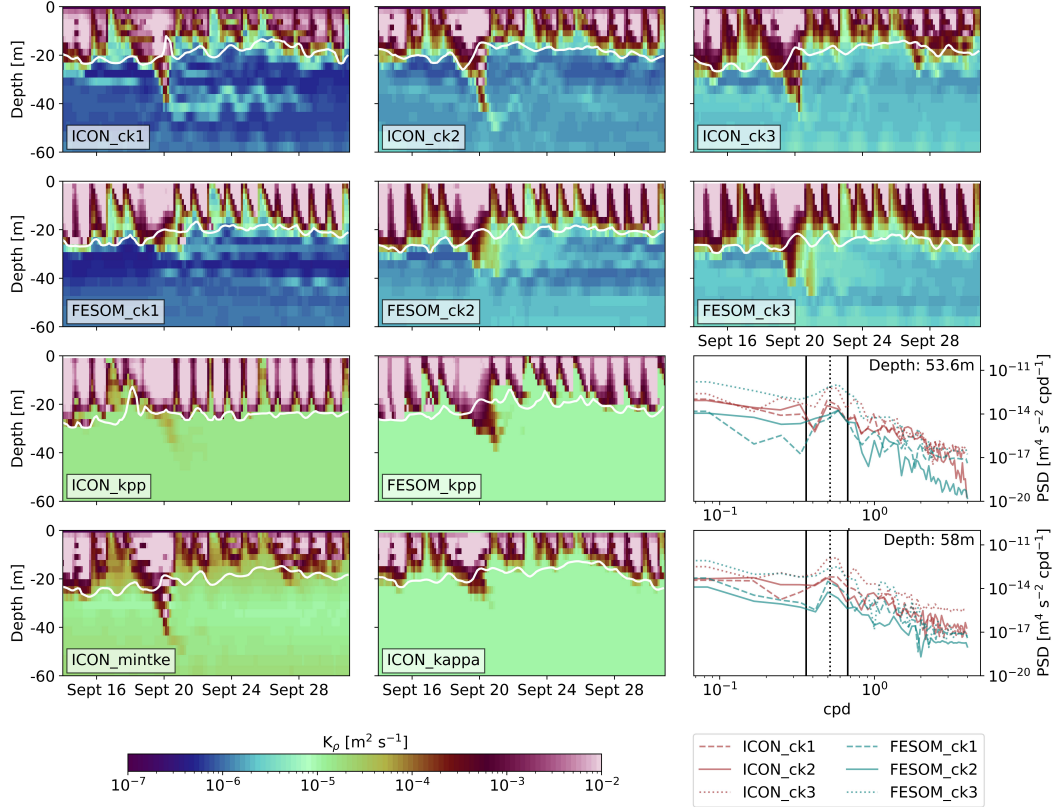


Figure 6. Snapshots of eddy diffusivities at the 15°N site in the models. The snapshots reveal the deep mixing caused by the storm, and the subsequent near-inertial diffusivity oscillations below the mixed layer. The depth of the mixed layer is indicated with the white line. The spectra of diffusivity time series between September 19th and 31st, 2015, at depths 53.6m and 58m confirm the presence of diffusivity oscillations at the near-inertial frequencies. The dotted lines mark f , and the two black solid lines envelop the range from $0.7f$ to $1.3f$.

and the vertical temperature gradient is sensitive to the mixing scheme choice; at 15°N , the mean NIKE bias is strictly negative, the NIWs are in phase, and the vertical temperature gradient is not highly sensitive to the mixing scheme choice. We now consider the NIKE biases at the two sites throughout the entire simulation period to illustrate what might be the cause of these differences.

We compare the probability density function (PDF) of the wind vector components and the wind speed, as well as the PDF of the relative bias between ERA5 output and PIRATA measurements. Throughout 2014 and 2015, the wind speed is systematically underestimated by ERA5. The magnitude of the bias is 7.88% at 11.5°N and 2.92% at 15°N . The PIRATA wind speed sensors have the precision of 2-3%. We therefore believe that the agreement between PIRATA and ERA5 simulated winds is good enough for NIW analysis in 2014 and 2015.

We define a "NIW event" in the time series of the bandpassed PIRATA current signal as the period in which NIKE is continuously above 25 Jm^{-3} for more than one inertial period. All 2014 and 2015 NIW events take place during boreal summer and fall, and in total there are nine of such events at 11.5°N and only one at 15°N . The probability distributions of relative NIKE biases are calculated based on data from periods when

NIWs are present at the sites: between June and November at 11.5°N, and between August and October at 15°N. While the PIRATA observations only show one NIW event at 15°N in 2015, the data from 2014 is also included, as in some of the TKE simulations NIKE is higher than 25 Jm⁻³ during that period.

Figure 7 shows that NIKE bias is sensitive to the mixing scheme choice at 11.5°N. The sensitivity is significantly reduced at the 15°N site (Fig. 8). The relative bias is defined as:

$$\text{Relative bias} = \frac{\text{NIKE}_{sim} - \text{NIKE}_{obs}}{\text{NIKE}_{obs} + c} \cdot 100\%, \quad (5)$$

where NIKE_{obs} is the observed (PIRATA) NIKE, NIKE_{sim} is the simulated NIKE, and the constant $c = 25 \text{ Jm}^{-3}$ is added to handle NIKE values that are close to zero. On the figure, the distribution of the relative bias is shown, together with the mean bias and its standard deviation. At 11.5°N, all TKE models overestimate the observed NIKE, while KPP underestimates it. The bias is sensitive to the TKE parameterization: when a higher percentage of TKE is used for mixing, the NIKE bias is reduced. The mean NIKE bias is smallest in the KPP simulation.

The distribution of the relative NIKE bias is also sensitive to the vertical mixing scheme at 15°N, but the sensitivity is not consistent with 11.5°N. Similarly to 15°N, Icon_ck1 simulates the largest positive bias and largest bias overall, and Icon_kpp is the only simulation which systematically underestimates NIKE. However, apart from Icon_ck1 and Icon_mintke, the average NIKE bias at 15°N is less than 3.5% across the sensitivity experiments. The 15°N site is therefore fundamentally different than 11.5°N during the simulation period: at 15°N, the major source of near-inertial energy is the tropical storm

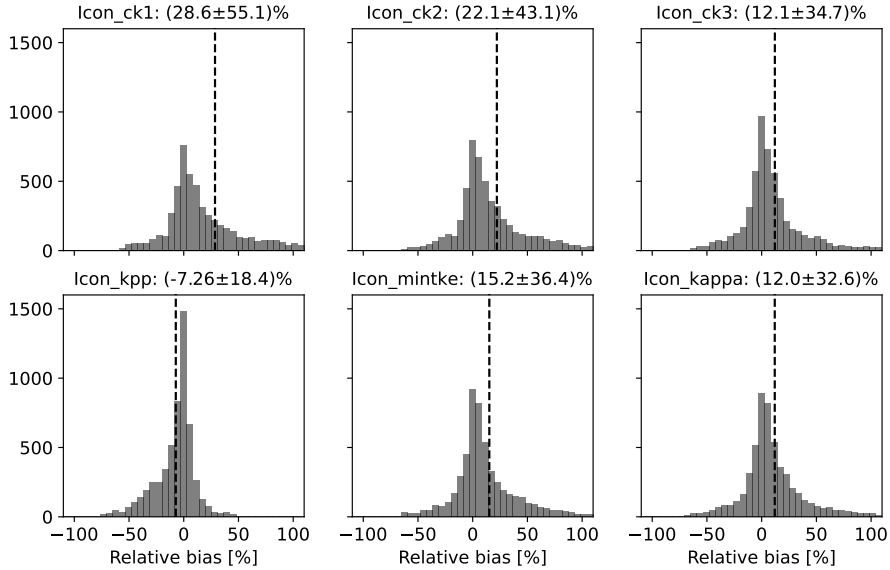


Figure 7. Probability distribution of NIKE relative bias in the ICON-O sensitivity studies between June and November in the years 2014-2015 at 11.5°N. The black dashed line indicates the mean bias, and the standard deviation of the bias is given in figure titles. Positive bias means that simulated NIKE is higher than observed. For example, in Icon_ck1, the systematic NIKE bias is 28.6%. In this simulation, 68% of the NIKE values in the time series are at minimum 26.5% too weak and at maximum 83.7% too strong compared to observations.

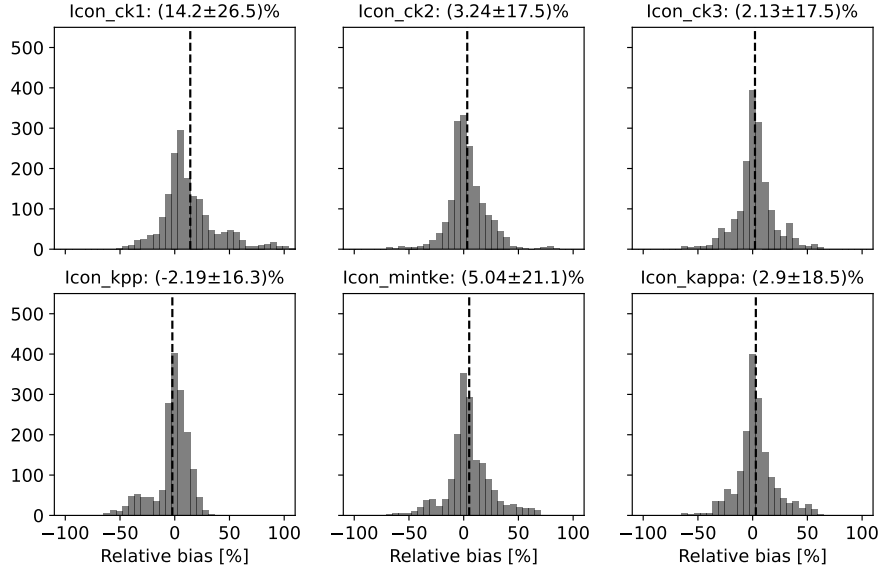


Figure 8. Probability distribution of NIKE relative bias in the ICON-O sensitivity studies between August and October of the years 2014 and 2015 at 15°N. The black dashed line indicates the mean bias, and the standard deviation of the bias is given in figure titles. Positive bias means that simulated NIKE is higher than observed. At this site, only one NIW event is detected. The time series is therefore dominated by near-zero NIKE values.

Ida. The NIWs at 11.5°N are forced by weaker but more frequent bursts of rotating winds, leading to generally higher near-inertial energy.

We found no statistically significant correlation of the NIKE bias with the wind speed, the MLD bias, or the stratification at the base of the mixed layer, the latter defined as the maximum N^2 value along the vertical coordinate. Hence, the contrasting characteristics of the modelled NIWs most likely stem from the combination of the physical conditions at the sites. At 11.5°N, the high density gradient across the ML (Table 2) produces strong near-inertial currents. NIW amplitude is inversely proportional to the depth of the mixed layer (D’Asaro, 1985). The strength of the near-inertial currents is therefore more sensitive to MLD biases when the ML is thin. At 15°N, the density gradient is lower and the ML deeper, hence the inertial current sensitivity to MLD biases is reduced.

4 Discussion

We present an evaluation of NIW-induced mixing at two sites in the tropical Atlantic in a range of sensitivity experiments using two different vertical mixing schemes: the Turbulent Kinetic Energy closure and the K-profile parameterization. We demonstrate that the eddy-resolving ocean models can simulate NIWs in the tropics to an extent, but the modelled near-inertial currents are too weak and the NIW-induced mixing underestimated. Out of the two parameterizations tested, KPP can model the observed mixing profiles better. At the 11.5°N site, ICON_kpp excels especially at simulating the depth of the mixed layer (Fig. 2c) and mean vertical temperature profile (Fig. 4f). Both KPP runs model the vertical stratification profile well (Fig. 5). The analysis

of NIKE biases at 11.5°N also shows that KPP is least biased in simulating the magnitude of the near-inertial currents (Fig. 7).

The models do not reproduce the enhanced diffusivities below 20m at the 11.5°N site (Fig. 5). However, at the 15°N site, mixing below the ML occurs (Fig. 6). The modelled and observed NIWs are in phase at the surface. In the models, the NIWs radiate down the water column and produce shear which induces mixing below the ML. The near-inertial oscillations in diffusivity persist below the ML for ten days after the storm passes by the mooring. The local winds are weak during those remaining days of September (Fig. 4h), suggesting that the interior near-inertial motions originate from the storm forcing. Hence, NIW-induced mixing could be a significant contributor to the local vertical turbulent cooling (see Foltz et al. (2020) for a detailed discussion about the seasonal cycle of the mixed layer heat budget at this site).

The TKE scheme across all sensitivity runs overestimates the stratification within the ML, which traps the NIWs within it and suppresses the mixing below it. ICON_ck3 is the only run which shows enhanced shear below the ML at 11.5°N. The enhanced TKE allows for vertical turbulent flux of momentum, while the thin ML produces relatively strong NIWs, but the shear remains too low to overcome the stratification and induce mixing (Fig. 5). While ICON-O and FESOM differ significantly across the tested parameters, they consistently show the same discrepancies between the mixing schemes: TKE simulates shallower mixed layer, lower ML diffusivities, and higher ML N^2 and Sh^2 compared to KPP and observations. Mean NIKE bias at 11.5°N is sensitive to the TKE scheme parameter c_k , which determines the fraction of turbulent kinetic energy allocated for vertical mixing. In the prognostic TKE equation, the eddy viscosity K_m also scales the vertical flux of shear and TKE. Hence, lower c_k values result in stronger stratification, which increase the inertial current velocity.

The TKE scheme is based on the 2.5 level turbulence closure model by Mellor and Yamada (1982). In order to simplify the scheme, they neglect the material derivative and the dissipation of the temperature variance. It is justified for stable and slightly unstable flows, but is likely to introduce numerical errors in highly unstable conditions. Fan et al. (2021) use the 2.5 level Mellor-Yamada scheme in the Navy Coastal Ocean Model (NYCOM) and find that NIW mixing is under-represented when compared to large eddy simulation (LES) results. They speculate that the highly unstable conditions induced by strong cooling of the mixed layer are the dominant cause for the scheme's inability to reproduce the NIW-induced mixing. Our results are consistent with theirs, indicating that the TKE scheme might be subject to the same limitation.

In this work, our assumption is that the NIW-induced mixing is not properly simulated because of the weaknesses of the mixing schemes. Other sources of biases that were not considered in this study might contribute to misrepresentation of NIWs in the models. Fan et al. (2021) find that turbulent mixing is better represented in the NYCOM model when the resolution at the mixed layer base is as high as in the surface layers. While this does not directly affect the NIW-induced mixing in their case, a finer resolution at the mixed layer base in our experiments could have countered biases in stratification (Table 2) and thereby indirectly countering the biases in NIW characteristics. Horizontal resolution might also play a role, as is explored in a study by Lévy et al. (2010). In an idealised basin, they find that increasing horizontal resolution in the NEMO model decreases the depth of the mixed layer. Aside from the resolution, non-local model biases such as advected offsets in temperature and salinity might have contributed to the suppression of NIW-induced mixing.

We demonstrate that near-inertial wave observations can be used to identify limitations in the performance of vertical mixing parameterizations. The presence of deep near-inertial oscillations following the storm at 15°N confirm the potential of high res-

olution simulations for exploring the dynamics of NIW-induced mixing in the ocean interior.

5 Conclusions

Two vertical mixing schemes (KPP and TKE) in two different models (ICON-O and FESOM) are evaluated on their ability to reproduce NIW characteristics and NIW-induced mixing at two PIRATA mooring locations (11.5°N and 15°N) in the tropical Atlantic. The main findings of our study are:

- Microstructure measurements show enhanced diffusivities below the mixed layer after the passing of a near-inertial wave at 11.5°N. The enhanced diffusivities are absent in the models regardless of the mixing scheme used.
- NIKE at both sites is sensitive to the mixing scheme choice. The sensitivity is higher at 11.5°N due to high stratification and shallow ML. KPP systematically underestimates the observed NIKE, while TKE systematically overestimates it.
- The KPP scheme simulates the vertical stratification profile and the depth of the mixed layer at 11.5°N more realistically than TKE. NIKE bias at 11.5°N is also generally smallest in the KPP runs.
- The models simulate NIW-induced mixing following a storm at 15°N. The near-inertial oscillations remain below the ML for 10 days after the storm passes.

Open Research Section

The M119 cruise data are available at <https://doi.pangaea.de/10.1594/PANGAEA.877375> (vmADCP; Brandt et al., 2017) and <https://doi.pangaea.de/10.1594/PANGAEA.920592> (microstructure; Fischer, 2020). The PIRATA data can be accessed at <https://www.pmel.noaa.gov/tao/drupal/disdel/>. The ePIRATA dataset is available at <https://www.aoml.noaa.gov/phod/epirata/> (Foltz et al., 2018). The ERA5 reanalysis product can be found at <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=overview> (Hersbach et al., 2020). The IFREMER MLD climatology is available at <https://cerweb.ifremer.fr/deboyer/mld/home.php> (de Boyer Montégut et al., 2004). Model output from ICON-O and FESOM used in this study is available at https://sid.erda.dk/cgi-sid/ls.py?share_id=DDGK0EDcTu.

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